Integrated System for Stress Monitoring and Fall Detection

Stephen Taiwo • Nikolaos Tsantekidis Supervisor: Brian Deegan

Ollscoil na Gaillimhe
University of Galway

1. Introduction

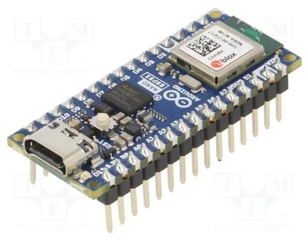
This project presents a wearable prototype that combines physiological and motion sensing to monitor user wellbeing in real time. The system is designed around a low-cost embedded platform and focuses on fall detection as the core feature, with stress-related monitoring explored through additional physiological signals.

2.Goals & Objectives

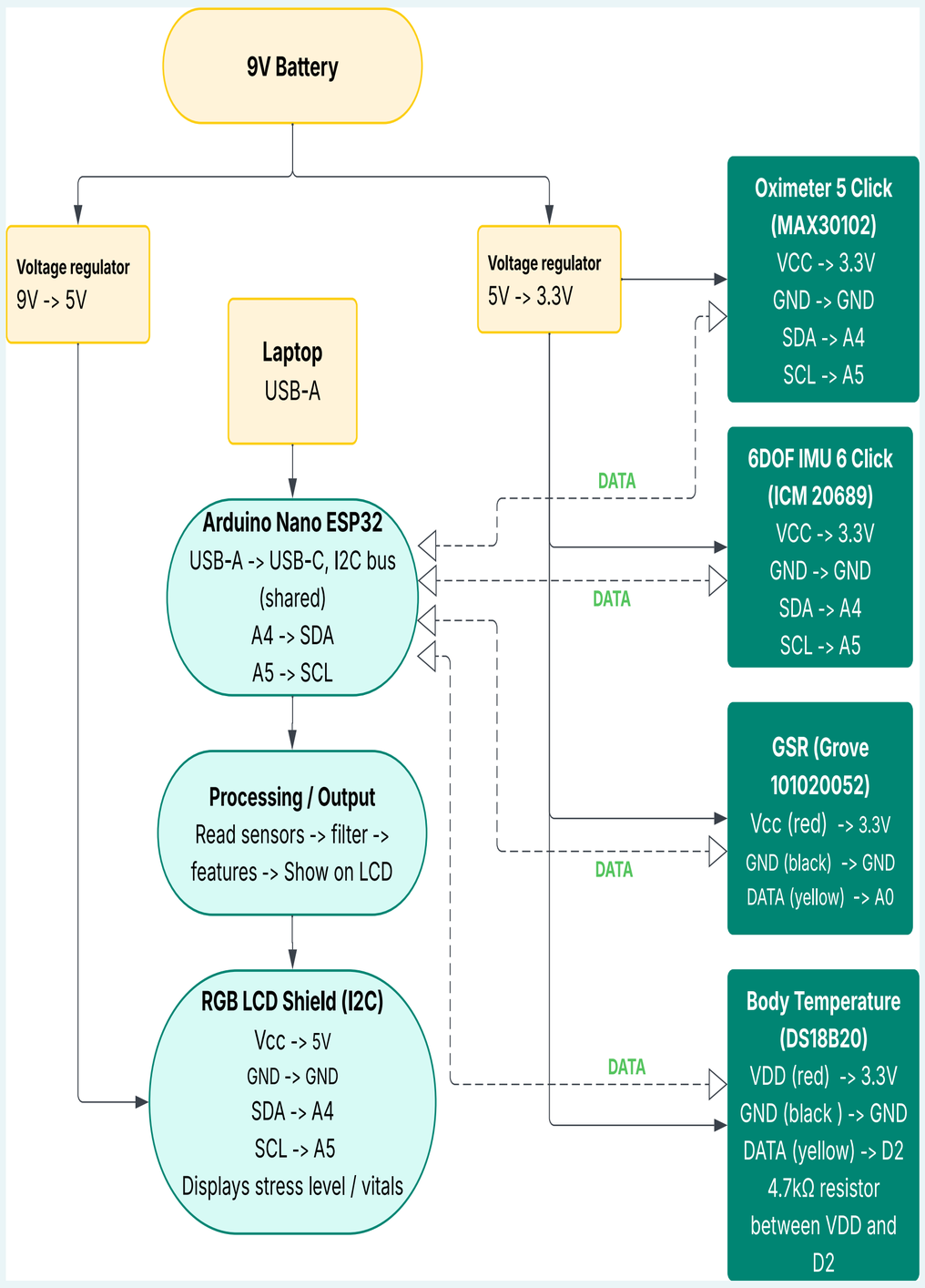
- Develop a multi-sensor prototype
- Measure heart rate, SpO_2, motion, skin conductance, and body temperature
- Implement a reliable fall detection algorithm
- Display live status locally and support real-time debugging

3.Hardware Platform

- **Arduino Nano ESP32** as the main controller
- **MAX30102** for heart rate and SpO_2
- **ICM20689 6DOF IMU** for fall detection
- **Grove GSR** for skin conductance
- **DS18B20** for body temperature
- **RGB LCD shield** for local feedback
- 9V Battery-powered with 5 V and 3.3 V regulators



4.System Architecture



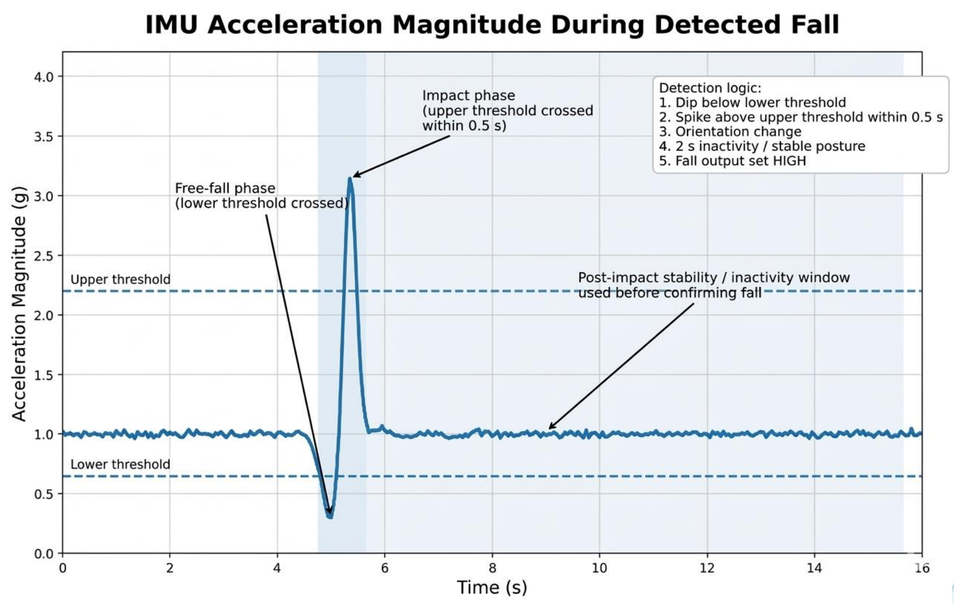
High-level system architecture showing power regulation, Arduino Nano ESP32, sensor interfaces, processing, and LCD output.

5.Fall Detection Method

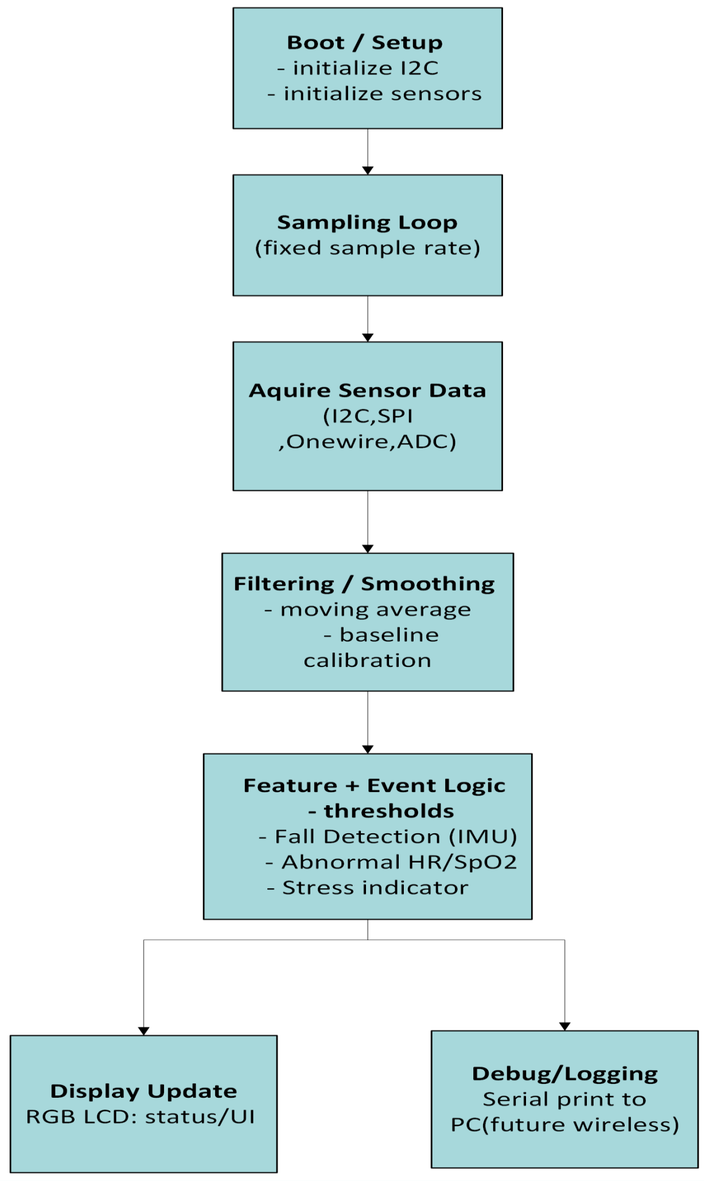
The implemented fall detection algorithm uses:

- Acceleration magnitude thresholds
- Orientation change
- A short inactivity window after impact

This state-machine approach reduces false alarms from normal movement while remaining suitable for embedded real-time execution



6.System Processing



The ESP32 acquires sensor data, applies filtering and threshold-based analysis, and detects fall events and physiological changes. Results are output via serial and displayed on the LCD.

7.Prototype Results

- IMU data acquisition was successfully validated for fall detection
- Heart rate and SpO2 readings were obtained reliably from MAX30102
- GSR and temperature sensing were functionally verified
- Serial monitoring and LCD output were integrated for live testing

8.Conclusion

The project demonstrates that a low-cost wearable platform can support real-time multi-sensor monitoring with fall detection and stress monitoring as primary features. Future work includes improving system efficiency for lower power operation and developing a compact enclosure design with wireless connectivity.